

UHLENBECK'S THEOREM II

IOANNIS KARAGIORGIS

ABSTRACT. We sketch the proof of Uhlenbeck's gauge fixing theorem, modulo boundary condition stuff. Main sources used are [DK90] and https://math.umd.edu/~bzh/MSRI_2021/main.html.

Recall the setup: let X^4 be closed and oriented and let P be a principal bundle over X . We are looking at the moduli space of ASD connections on X that live in a suitable Sobolev space. When we say a connection A is in L_k^p , we mean its local matrix representations are in L_k^p . Write $\mathcal{A}_k^p$ for the space of L_k^p connections. Recall that this is an affine Banach space on $T^*X \otimes \text{ad } P$, and is acted upon by the gauge group $\mathcal{G}_{k+1}^p$. The moduli space is

$$\mathcal{M}_k^p = \{A \in \mathcal{A}_k^p : F_A^+ = 0\} / \mathcal{G}_{k+1}^p.$$

What we want: for suitable k and p , if $A \in \mathcal{A}_k^p$ and $F_A^+ = 0$, then A is gauge equivalent to a smooth connection. But we immediately run into the following problem. In local coordinates, $F_A = dA + A \wedge A$, so

$$F_A^+ = \underbrace{d^+ A}_{\text{linear part}} + \underbrace{(A \wedge A)^+}_{\text{non-linear part}}.$$

The issue is that even the linear part of this equation is not elliptic. Indeed,

$$d^+ : \underbrace{\Omega^1(X) \otimes \mathfrak{su}(2)}_{\dim_{\mathbb{R}}=12} \rightarrow \underbrace{\Omega_+^2(X) \otimes \mathfrak{su}(2)}_{\dim_{\mathbb{R}}=9},$$

so just looking at the dimensions, we see that the symbol will always have kernel. Under the action $\mathcal{G}_{k+1}^p \curvearrowright \mathcal{A}_k^p$, we could start with a smooth solution and apply a non-smooth gauge transformation to it to obtain a non-smooth solution. Uhlenbeck's gauge fixing theorem tells us that we can choose a nice gauge representative from each orbit. To do so, we need to throw in an additional equation, which together with the equation $d^+ A = 0$, will form an elliptic system.

Theorem 1 (Uhlenbeck). *There exist constants $\varepsilon_0 > 0$ and $C > 0$ such that for any L_1^2 -connection A on (the trivial bundle of) B^4 with $\|F_A\|_{L^2(B^4)} < \varepsilon_0$, there exists $g \in \mathcal{G}_2^2(B^4)$ such that*

- (1) $d^*(g(A)) = 0$;
- (2) $\|g(A)\|_{L_1^2} \leq C\|F_A\|_{L^2}$.

Remark. This is a weaker version of the full theorem (we are excluding the boundary condition), but it is sufficient for our purposes.

What the theorem is saying: assuming the curvature of A is small (in L^2 norm), we can find a gauge representative A' of the equivalence class of A in the moduli space such that A' satisfies the *gauge fixing equation* $d^* A' = 0$ (we say A' is a *Coulomb gauge representative*). The upshot is that now the linear part of the equations $d^+ A + (A \wedge A)^+ = 0$ and $d^* A = 0$ is elliptic, i.e.

$$\begin{cases} d^+ A = 0 \\ d^* A = 0 \end{cases}$$

is the elliptic system we want. Indeed, we can verify that $d^+ + d^* : \Omega^1 \rightarrow \Omega_+^2 \oplus \Omega^0$ is elliptic by computing its symbol.

Remarks.

- (1) The theorem gives a Coulomb gauge on B^4 , but we can use it for any compact 4-manifold by first applying it to each ball in an open cover of the manifold and then patching all of the gauge transformations together. This does require dealing with boundary stuff though.

- (2) This is not special to $SU(2)$; it works for any choice of *compact* Lie group as the structure group.
- (3) The constants ε_0 and C that appear in the theorem depend on the chosen metric on B^4 .
- (4) Because B^4 is contractible, every principal bundle over it is trivial. So there is no loss of generality in the theorem.

Proof. The first step is to find a family of connection matrices $A(t)$, for $t \in I = [0, 1]$, such that

- (1) $A(0) = 0$ (zero matrix);
- (2) $A(1) = A$;
- (3) $\|F_{A(t)}\|_{L^2(B^4)} < \varepsilon_0$.

The result of the theorem clearly holds for the zero connection. So at $t = 0$ we start with the zero connection and $g = \text{id}$, and then we try to extend to get a gauge transformation at $t = 1$. Let $\delta_t: \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be the dilation map $\delta_t(x) = tx$ for $t \in I$, and define $A(t) = \delta_t^*(A)$ over B^4 . We are basically just scaling the domain to get a path from the given connection matrix A to 0. As a consequence of the conformal invariance of the L^2 norm of curvature, we have

$$\int_{B^4} |F_{A(t)}|^2 d\mu = \int_{|x| \leq t} |F_A|^2 d\mu \leq \int_{B^4} |F_A|^2 d\mu,$$

from which we see that the L^2 norm of the curvature for each connection on the path $A(t)$ is controlled by that of A .

The second step is to somehow pull back the family $A(t)$ to S^4 , and reduce the problem to an easier one, by avoiding having to deal with a boundary value problem over B^4 (plus analysis is easier on compact manifolds). Start with identifying B^4 with the northern hemisphere in S^4 . Let $r: S^4 \rightarrow S^4$ be the reflection map that is the identity on the equatorial S^3 and interchanges the two hemispheres. Let $p: S^4 \rightarrow B^4$ be the map that is the identity on the northern hemisphere (the B^4 we are working on) and equal to r on the southern hemisphere. Now take pull-backs $p^*(A(t))$ to get a family of connections on S^4 , i.e. ‘double’ each connection. The issue here is that p is not differentiable on the equator (it is a Lipschitz map, as it is a.e. differentiable with bounded derivative). We at least have that $p^*(A(t))$ is an L^∞ 1-form on S^4 and its curvature an L^∞ 2-form on S^4 and

$$\int_{S^4} |F_{p^*(A(t))}|^2 d\mu = 2 \int_{B^4} |F_{A(t)}|^2 d\mu.$$

To deal with the fact that the connections $p^*(A(t))$ are not smooth, we need to mollify p . Choose a family of smooth maps $p_\varepsilon: S^4 \rightarrow B^4$ with uniformly bounded derivatives that converge uniformly to p as $\varepsilon \rightarrow 0$ and with each p_ε equal to p outside the ε -neighbourhood of the equatorial S^3 . For each t , $p_\varepsilon^*(A(t))$ is smooth and we have

$$\int_{S^4} |F_{p_\varepsilon^*(A(t))}|^2 d\mu \leq 2 \int_{B^4} |F_{A(t)}|^2 d\mu + \varepsilon C \|F_{A(t)}\|_{L^\infty},$$

for some constant C . Thus, to prove the theorem for a given connection A , we choose ε sufficiently small, and show there is a constant $\zeta > 0$ such that if $\|F_{p_\varepsilon^*(A(t))}\|_{L^2} < \zeta$ for all t , then we can get a Coulomb gauge representative for $p_\varepsilon^*(A)$. We then choose the constant ε_0 appearing in the theorem to be $\varepsilon_0 < 2^{-1/2}\zeta$. Restricting back to B^4 , we get a Coulomb gauge representative for A over $B^4(1 - \varepsilon)$. Finally, to get the Coulomb gauge we want over the whole B^4 , we let $\varepsilon \rightarrow 0$. There are some technicalities here, but we don’t need to worry about them for the remainder of the proof. In what follows, we can just assume we are working on $B^4(1 - \varepsilon)$ for $0 < \varepsilon \ll 1$.

From now on, we abuse notation and simply write $A(t)$ for the pull-backs. The $A(t)$ are a continuous one-parameter family of L^2_1 -connections on (the trivial bundle of) S^4 such that $\|F_{A(t)}\|_{L^2} < \zeta$ and $A(0) = 0$. We wish to show that for each $t \in I$, there exists $g_t \in \mathcal{G}_2^2$ such that

$$d^*(g_t(A(t))) = 0 \tag{1}$$

and

$$\|g_t(A(t))\|_{L^2_1} \leq C \|F_{A(t)}\|_{L^2}. \tag{2}$$

To extend from $t = 0$, we use a continuity argument. Let

$$S := \{t \in I : g_t \text{ exists satisfying 1 and 2}\}.$$

Note that S is non-empty, since $0 \in S$. We will prove that S is both open and closed, thus proving $S = I$. Then, in particular $t = 1 \in S$, which yields the result.

We first show that S is closed. Let (t_i) be a sequence in S and suppose $t_i \rightarrow t$. We will show that (g_{t_i}) has a weak limit in L_2^2 . Write $g_i = g_{t_i}$, $A_i = A(t_i)$, $A = A(t)$ and $B_i = g_i(A_i)$. By definition of the gauge group action, locally we have

$$\begin{aligned} B_i &= A_i - (d_{A_i} g_i) g_i^{-1} \\ &= A_i - [A_i, g_i] g_i^{-1} - (dg_i) g_i^{-1} \\ &= g_i A_i g_i^{-1} - (dg_i) g_i^{-1}, \end{aligned}$$

hence

$$dg_i = g_i A_i - B_i g_i.$$

Because $SU(2)$ is compact, we get an L^∞ bound for free on the g_i , i.e. $\|g_i\|_{L^\infty} \leq C$. The B_i and A_i being bounded in L_1^2 implies they are bounded in L^4 using Sobolev embedding, and we get $\|dg_i\|_{L^4} = \|g_i A_i - B_i g_i\|_{L^4} \leq C$, thus $\|g_i\|_{L_1^4} \leq C$. Also, $B_i g_i \in L_1^2 \times L_1^4 \subset L_1^2$ by Sobolev multiplication, so $\|B_i g_i\|_{L_1^2} \leq C$ and similarly $\|g_i A_i\|_{L_1^2} \leq C$. So $\|dg_i\|_{L_1^2} \leq C$, thus $\|g_i\|_{L_2^2} \leq C$. As the g_i are uniformly bounded in L_2^2 , there exists a subsequence g_{i_k} such that $g_{i_k} \rightharpoonup g$ in L_2^2 (thus strongly convergent in L_1^2). The weak limit in L_2^2 becomes a strong limit when we pass to L_1^2 . Together with uniqueness of weak limits, we get $g_i \rightarrow g$ in L_1^2 .

By assumption, we have $A_i \rightarrow A$ in L_1^2 . We also have the uniform bound $\|B_i\|_{L_1^2} \leq C\zeta$, so there exists a subsequence B_{i_k} such that $B_{i_k} \rightarrow B$ in L_1^2 . Putting everything together, and using Sobolev multiplication, we get $g(A) = B$. We argue similarly to show that both conditions 1 and 2 are satisfied. In other words, $t \in S$.

We now show that S is open. We start with a connection A that satisfies equation 1 and show that any small perturbation $A + a$ can also be gauge fixed. Here, a is a matrix of 1-forms with small norm. Consider the perturbation map $\Phi: L_2^2(\text{ad } P) \times L_1^2(T^*S^4 \otimes \text{ad } P) \rightarrow L^2(\text{ad } P)$ given by

$$(\xi, a) \mapsto d^*(\exp(\xi)(A + a)).$$

Here $A = A(t)$ and we assume WLOG that $g(t) = \text{id}$ (otherwise we can just compose with $g(t)$ locally). Look at the linearisation of this perturbation at zero:

$$d\Phi_{(0,0)}(\xi, a) = d^*([\xi, A] + a - d\xi).$$

We must show that for all a there exists ξ such that $d^*([\xi, A] + a - d\xi) = 0$. This actually follows from the fact that on a *closed* manifold, $\text{im}(d^*) = \text{im}(d^*d)$ ¹. A perturbation of a surjection is still a surjection, so for A small, we have

$$\text{im}(d^*d) = \text{im}(d^*([- , A] - d)).$$

By some analysis results for Banach manifolds, this implies there exists $g_{t'}$ such that $d^*(g_{t'} A(t')) = 0$, where $A(t')$ is a small perturbation of A .

It remains to prove that inequality 2 is also an open condition. That is, the inequality still holds after applying a small perturbation. The idea here is to use a “rearranging argument” and the basic elliptic estimate. Write $B(t) = g_t(A(t))$. We have $F_B = dB + B \wedge B$ and we assume $d^*B = 0$, so rearranging we get $(d + d^*)B = F_B - B \wedge B$. Applying the basic elliptic estimate to the elliptic operator $d + d^*$, we get

$$\|B\|_{L_1^2} \leq c_1(\|F_B\|_{L^2} + \|B \wedge B\|_{L^2}).$$

¹Hodge decomposition gives $\Omega^2 = \mathcal{H} \oplus \text{im}(d) \oplus \text{im}(d^*)$, where the harmonic forms are $\mathcal{H} = \ker(d) \cap \ker(d^*)$. Applying d^* to $\mathcal{H}$ we get zero by definition. If we apply d^* to $\text{im}(d^*)$ we also get zero because $(d^*)^2 = 0$. Therefore, $\text{im}(d^*d) = \text{im}(d^*)$.

Note that this is the improved version of the estimate because $H^1(S^4) = 0$, so Hodge decomposition gives $\Omega^1(S^4) = \text{im}(d) \oplus \text{im}(d^*)$ (see (A8) Remark (iii) in [DK90]). By Sobolev multiplication, $\|B \wedge B\|_{L^2} \leq c_2 \|B\|_{L^4} \|B\|_{L_1^2}$. If $\|B\|_{L_1^2} < 1/(2C)$, then

$$(1 - C\|B\|_{L_1^2})\|B\|_{L_1^2} \leq C\|F_B\|_{L^2},$$

which gives

$$\|B\|_{L_1^2} \leq 2C\|F_B\|_{L^2}.$$

□

REFERENCES

- [DK90] S. K. Donaldson and P. B. Kronheimer, *The geometry of four-manifolds*, Oxford Mathematical Monographs, The Clarendon Press, Oxford University Press, New York, 1990. Oxford Science Publications. MR1079726